Design and Analysis of a Pin-Fin Heat Sink for Smartphone Cooling

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Abstract

Smartphones operating under sustained computational loads, such as navigation applications in high ambient temperatures, are susceptible to overheating. This report investigates the feasibility of using a pin-fin heat sink attached to the rear surface of a smartphone to reduce operating temperature. Analytical fin theory is applied to evaluate the effects of fin material, fin geometry, and array configuration on heat dissipation performance.

1 Introduction

Modern smartphones generate significant heat during extended use. Excessive temperature rise can degrade performance, reduce battery life, and cause user discomfort. Passive cooling solutions such as pin-fin heat sinks offer a low-power method to enhance convective heat transfer.

This study analyzes a pin-fin heat sink design attached to the back of a smartphone and evaluates how fin length, material, and array configuration influence the phone surface temperature.

2 Physical Model and Assumptions

- Steady-state heat transfer
- Uniform base temperature across the phone surface
- Negligible thermal resistance between phone and fin base
- Adiabatic fin tips
- Constant convection coefficient
- Radiation effects neglected

Phone base area: A_b, heat dissipation rate: Q, ambient temperature: T_∞.

3 Analytical Formulation

Corrected fin length for adiabatic tip:

$$L_c = L + \frac{D}{4} \tag{1}$$

Fin parameter:

$$m = \sqrt{\frac{4h}{kD}} \tag{2}$$

Fin efficiency:

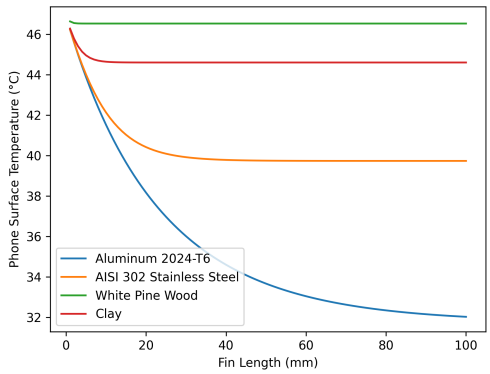
$$\eta_f = \frac{\tanh(mL_c)}{mL_c} \tag{3}$$

Base temperature:

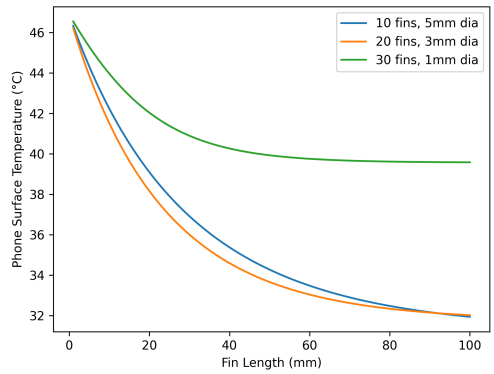
$$T_b = T_\infty + \frac{Q}{hA_{b,f} + N\eta_f hA_f} \tag{4}$$

4 Results and Discussion

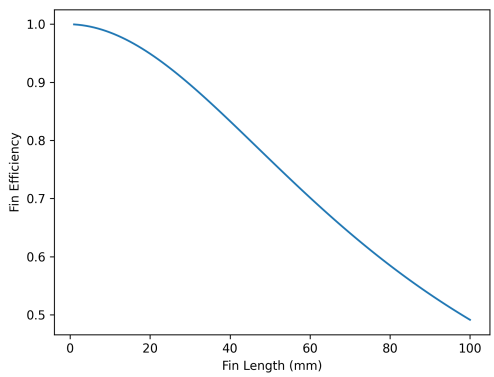
4.1 Parametric Studies



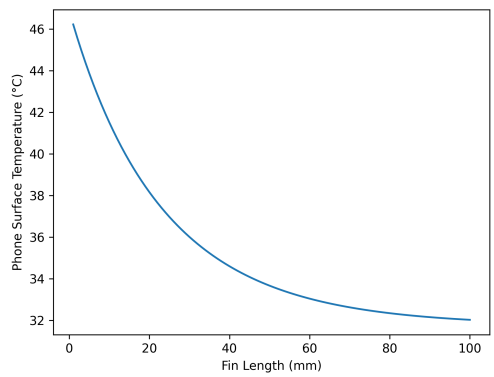
(a) Different materials



(b) Different fin arrays (Aluminum)



(c) Fin efficiency (optimal)



(d) Phone temperature (optimal design)

Figure 1: Parametric study of phone surface temperature and fin efficiency. All figures shown on one page.

5 Conclusions

Pin-fin heat sinks can significantly reduce smartphone surface temperature under steady-state conditions. High-conductivity materials such as Aluminum 2024-T6 perform best. The optimal design is 20 fins, 3 mm diameter, balancing heat dissipation and manufacturing constraints.

6 Appendix: Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Enable interactive plotting
plt.ion()

# Given constants
Q_total = 10.0          # W
T_inf = 20.0            # C
h = 50.0                # W/m^2*K
A_base = 0.06 * 0.124   # m^2

fin_lengths = np.linspace(0.001, 0.1, 100)

# Material properties
materials = {
    "Aluminum␣2024-T6": 177,
    "AISI␣302␣Stainless␣Steel": 15.1,
    "White␣Pine␣Wood": 0.11,
    "Clay": 1.3
}

# Fin arrays
fin_arrays = {
    "10␣fins,␣5mm␣dia": (10, 0.005),
    "20␣fins,␣3mm␣dia": (20, 0.003),
    "30␣fins,␣1mm␣dia": (30, 0.001)
}

def phone_temp(Q, k, L, h, N, D, A_base, T_inf):
    Lc = L + D/4
    m = np.sqrt(4*h/(k*D))
    eta = np.tanh(m*Lc)/(m*Lc)
    A_fin = np.pi * D * Lc
    A_base_free = max(A_base - N*np.pi*(D/2)**2, 0.0)
    T_base = T_inf + Q/(h*A_base_free + N*eta*h*A_fin)
    return T_base, eta

# 1) Phone temp vs fin length (materials)
array_choice = "20␣fins,␣3mm␣dia"
N_fins, D = fin_arrays[array_choice]

plt.figure()
for material, k in materials.items():
    temps = []
```

```
    for L in fin_lengths:
        T_base, _ = phone_temp(Q_total, k, L, h, N_fins, D, A_base, T_inf)
        temps.append(T_base)
    plt.plot(fin_lengths*1000, temps, label=material)
plt.xlabel("Fin␣Length␣(mm)")
plt.ylabel("Phone␣Temperature␣($^\circ$␣C)")
plt.legend()
plt.grid(True)
plt.savefig("figures/materials_vs_length.png", dpi=300)

# 2) Phone temp vs fin arrays
material_choice = "Aluminum␣2024-T6"
k = materials[material_choice]

plt.figure()
for array_name, (N_fins, D) in fin_arrays.items():
    temps = []
    for L in fin_lengths:
        T_base, _ = phone_temp(Q_total, k, L, h, N_fins, D, A_base, T_inf)
        temps.append(T_base)
    plt.plot(fin_lengths*1000, temps, label=array_name)
plt.xlabel("Fin␣Length␣(mm)")
plt.ylabel("Phone␣Temperature␣($^\circ$␣C)")
plt.legend()
plt.grid(True)
plt.savefig("figures/length_vs_phone_temp_arrays.png", dpi=300)

# 3) Optimal design
N_opt, D_opt = fin_arrays["20␣fins,␣3mm␣dia"]
k_opt = materials["Aluminum␣2024-T6"]

temps_opt = []
eff_opt = []

for L in fin_lengths:
    T_base, eta = phone_temp(Q_total, k_opt, L, h, N_opt, D_opt, A_base, T_inf)
    temps_opt.append(T_base)
    eff_opt.append(eta)

plt.figure()
plt.plot(fin_lengths*1000, temps_opt)
plt.xlabel("Fin␣Length␣(mm)")
plt.ylabel("Phone␣Temperature␣($^\circ$␣C)")
plt.grid(True)
plt.savefig("figures/temps_vs_length_optimal.png", dpi=300)

# 4) Fin efficiency
```

```
plt.figure()
plt.plot(fin_lengths*1000, eff_opt)
plt.xlabel("Fin␣Length␣(mm)")
plt.ylabel("Fin␣Efficiency")
plt.grid(True)
plt.savefig("figures/efficiency_vs_length.png", dpi=300)

input("Press␣Enter␣to␣close␣all␣plots...")
plt.close("all")
```